SCORE: A Framework for Synthesizing Complete, Owned, Responsive Environments

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Abstract

Games, films and robot training need 3D worlds that are complete, editable and owned by their makers; world models and one-shot generators give frames no engine opens, or one fused scene without separate objects or clear rights. Studios build worlds in stages, from concept art to review. SCORE keeps these stages and automates them with open vision foundation models, procedural tools and coding agents, while the creator chooses and reviews. Every stage is a node of one runner: its output is keyed on its inputs and code, judged by gates that answer pass, fail or unknown, and accepted only when nothing got worse, so a fault is fixed in its own stage and only what follows it runs again; paid steps are cleared against a budget read from the provider's bill. Every model allows commercial use of its output, so the creator stays the author. We describe the framework as built for its first world, the game world of 2099 with 28 places, and the measured gates of its surfaces, resting, light, motion, effects, sound and people. Code and paper are available at https://github.com/Babon-Innovations-b-v/score (MIT license).

1 Introduction

Interactive 3D worlds are the raw material of games, film and embodied AI [1], and building one is expensive: every object is modelled, surfaced, placed, given collision, lit and given sound in one style, and the world stays editable. Generative models promise to take over much of this, but in GDC's 2026 survey 52 percent of game workers thought generative AI was hurting the industry [2]; a tool for building worlds has to leave the result in the creator's hands.

World models generate a world as frames [3], [4] that no engine can open. 3D world generators [5], [6] produce geometry, but mostly as one fused asset, from which separate objects are recovered after the fact [7], [8]. Both work in one shot, so a creator cannot find or change the step behind a fault. Procedural systems [9], [10] and coding agents that write scenes as programs [11], [12], [13], [14] keep the world explicit, but rebuild a scene that already exists or draw their objects from a fixed library, and such agents judge their own geometry unreliably [15]. Open vision foundation models [16] now read shapes [17], masks [18], depth [19] and parts [20] out of pictures, and a coding agent can connect them as tools; none of them provides one style across objects that different models made, which the field names as an open problem [21], [22].

Studios build worlds in stages, each with an output a person can check; SCORE keeps these stages and automates them. Its worlds are **complete**: every object, surface, light and sound exists wherever the player looks. They are **owned**: every model in the framework allows commercial use of its output, so the creator stays the author, holding the full commercial rights. They are **responsive**: every object has real collision, mass and material. And they can be any kind of **environment**. To

our knowledge, no other openly available world generation work joins vision foundation models into engine-ready scenes that are built from reference pictures, steered by a person's choices at each step, and made only with models that allow commercial use of their output.

We contribute **the SCORE framework**, stages as nodes of one runner that keys, judges and accepts every output over one canonical scene (Section 2); **measured gates** for every layer of a place and a completion gate over them (Section 3); and **evidence from a first world**, with its spend per stage (Section 4).

2 The framework

A world is a table of stages; world 1's has 95 rows. A **node** is one stage for one place (`assembly@hangar`) or for the world, and one runner makes, judges and accepts every node. The creator's picks (concepts, style, references) are the inputs; from there everything runs without a typed yes, and the creator reviews.

2.1 Keys, gates and accepts

Keys. A node is made only when its key has no finish record. The key hashes the stage and place, a code version, the contracts of its output and inputs, its parameters, a seed derived from the world's, the pinned model images, every world file it reads and the accepted output of every node it reads. The code version hashes every source file the stage imports, leaving out the runner and the renting code, which change what a run costs, not what it makes. A fault fixed in one stage moves only its key and those after it. Outputs are kept by content hash, never overwritten.

Gates and accepts. Each gate answers pass, fail or unknown, with measures and a reason; missing evidence is unknown, and unknown blocks like fail. The runner accepts an output by itself when every gate passed and it is no worse than the one accepted before: no answer dropped, no measure moved the wrong way by a step a viewer could see, no gate missing. Later nodes read only accepted outputs. The creator keeps four commands, from the creator's own terminal: reject an accepted output (it moves back to the one before, and every node that read it is stale), accept a worse one with a reason, waive one gate for one key, and override the runner for one paid run. When the creator accepts a worse output, that accept records what was allowed, and a later output no worse than the one taken is accepted with that accept named. None is needed for a run; a barrier keeps agents from them (a terminal, a store that refuses the creator's records from an agent's process, and a hook).

Spending and review. A paid node is cleared only when every gate on its inputs passed and the month's ceiling has room for its estimate (Section 3.7); everything that rents a machine or calls a paid model refuses without that ticket, and buying again more than EUR 5 of accepted outputs needs a written reason. Each node's review page is built from the store alone, and the creator reviews a place in a USD viewer on the store's copy of its accepted assembly; nothing is rendered for review.

The creator steers: the ship. Figure 1 shows the creator's control on one piece of world 1, the ship. The creator's picks were found pictures of a shuttle, a one-deck cutaway, a cockpit and crew quarters; we do not show them. From them an agent planned the bridge alone (v1, a 4.6 by 4.6 m room of 27 pieces). The creator corrected the plan in words: a smaller cockpit, rooms the game needs, a long flat wedge rather than a rocket, a nose less pointed, and an outline measured from drawings rather than shaped from words. Each correction made a new plan, and the creator

accepted the seventh. Every plan is ours and cost nothing; the two drawings the outline was traced from cost EUR 0.28 and are not shown. The ship's shape follows found artwork the creator chose to use: shuttle concept by PAUL (zana), found online.

2.2 From concept to scene

Inventory, plan and record. A coding agent lists every visible element of the concept as an inventory row (kind, anchor, size, count) or drops it with a written reason; nothing is made or placed that is not a row. It draws a dimensioned plan, gated on `mapped` (every element a row, a door or a written drop; every door says where it leads). The scene record is written from the plan, inventory, made ground and sky, gated on `game-free` (the record cites nothing from the game the world came from), `plan-matched` (the shell within 5 cm of the plan), `openings` (every doorway onto a floor, no view into nothing) and `seed-matched` (every old entry matched or dropped with a reason). The `game-free` gate sees cited numbers only. On world 1 the records cite nothing, but in 18 of its 28 places the data they are written from still take positions from the source game (Section 4).

The canonical scene. Every place is one OpenUSD stage [23]; each object, one per real-world object, carries its kind, transform, collision and each part's library surface. Its layers lie in a fixed order: a slot for the creator's edit layer on top, then people, sound, light, motion, effects, crops, machines, settle, ground and the generated base. The edit layer exists only as that slot, an empty layer the exporter makes once and never writes again; a test on a two-box place shows an edit there surviving a regenerated base, but the runner's assembly does not yet carry a creator's edit from one assembly to the next. Each place is assembled once, from every layer's accepted output; its `layers` gate answers unknown while a needed layer has none, and an old output never stands in.

Models, generation first. Every object goes the prop route first: Nano Banana Pro [24] draws its close-up from its crop of the concept, Pixal3D [17] builds it, the surface stage paints it, and a likeness gate compares its outline from eight sides with its close-up. Only an object whose checks fail is built by a code builder a coding agent writes, reversing the earlier parts check (the creator's choice, 12 October 2026). On 13 of the hub's objects both orders end the same (Table 1, Figure 2), but generation first draws a close-up of every object. Every close-up of eight thin and flat objects failed the shape check, so code builds carry them. The hatch wheel shows the gain: its code build's outline overlap was 0.45, a fail, and its generated model's 0.60. Most of world 1's models predate the change and enter as unverified seeds.

Table 1. How 13 hub objects, chosen to cover the kinds, end in each order. Close-ups cost EUR 8.12 and the sample's prop route EUR 13.80, about EUR 2.20 an object; no take was baked (the run stopped at the keyboard's paint gate).

Order	Prop route	Code build	Needs a new builder
Generation first	4	7	2
Code first	4	7	2

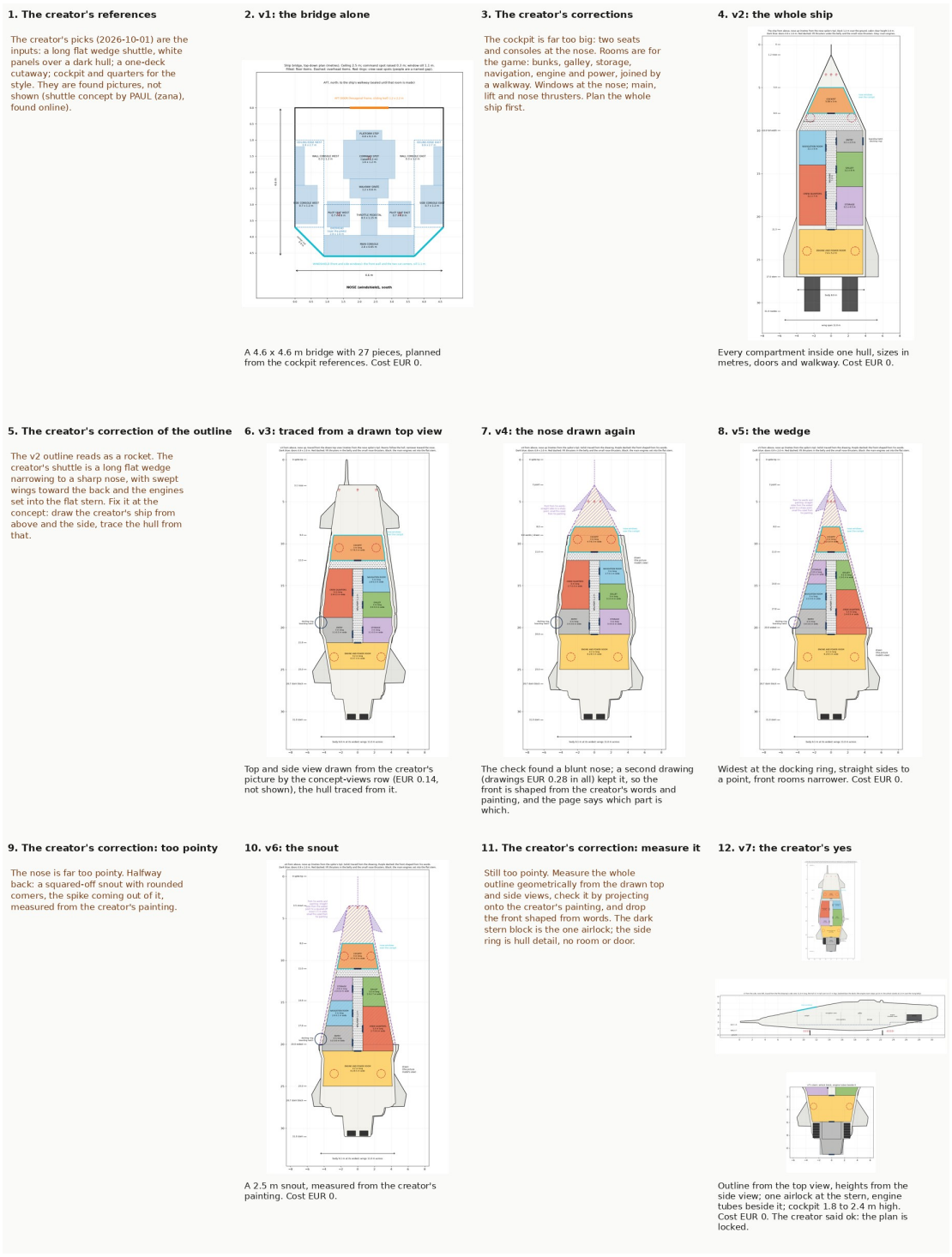


Figure 1: The creator steers the ship's plan, in twelve steps: the creator's picks, described in words (found pictures, not shown); plan v1 of the bridge alone; the creator's corrections; v2 of the whole ship; the creator's correction of the outline; v3 to v6; the creator's two corrections of the nose; v7, which the creator accepted. All plans are ours. Shuttle concept by PAUL (zana), found online.

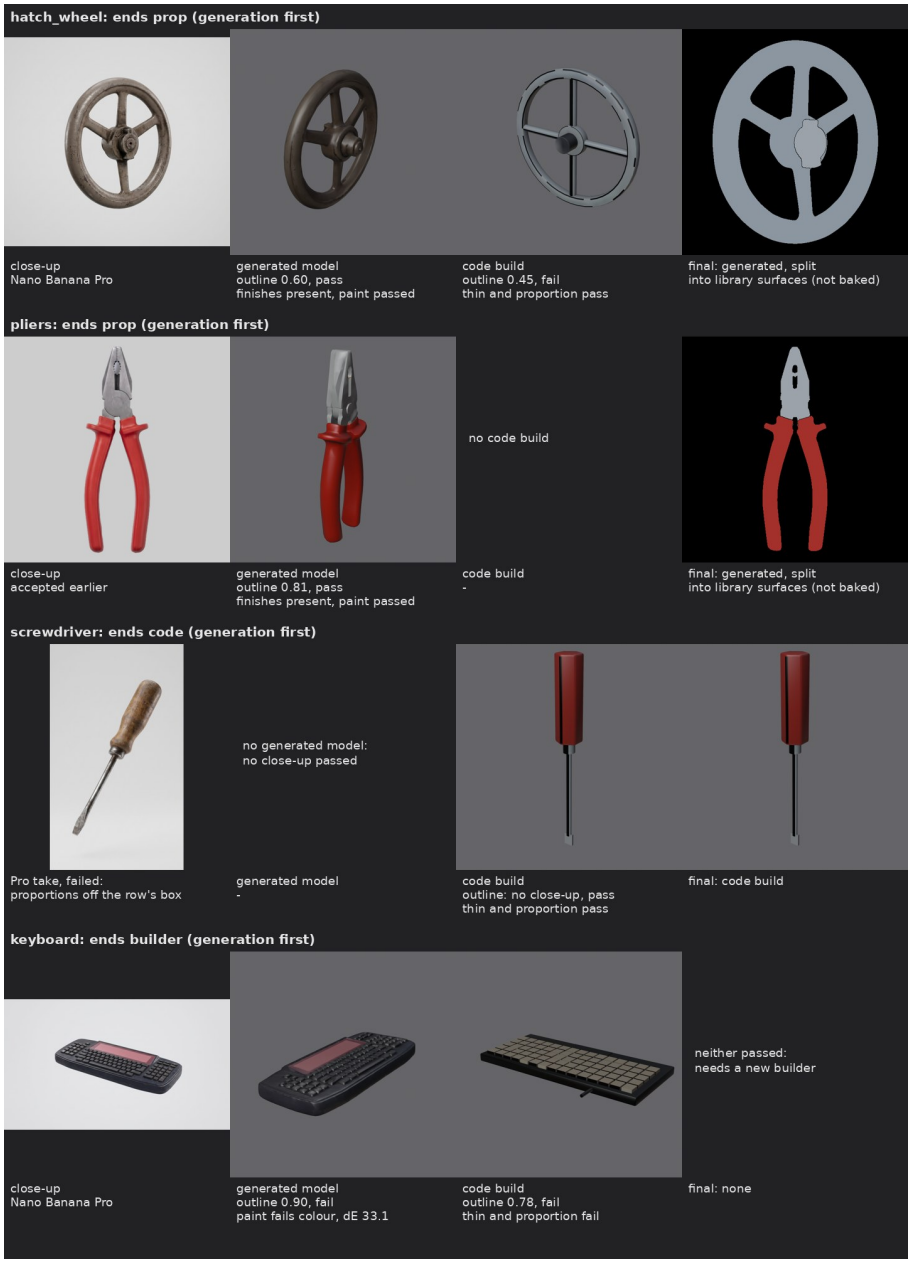


Figure 2: Generation first on four hub objects. Columns: close-up, generated model, code build, final model. The hatch wheel's generated model passes its outline check at 0.60 where its code build fails at 0.45; the pliers pass at 0.81; every close-up of the screwdriver failed, so its code build carries it; the keyboard fails both ways (paint ΔE 33.1; code build too thin and out of proportion) and needs a new builder. Generated models are split into library surfaces, not baked.

3 Stages and their gates

3.1 Surfaces: the finish map

Models give shape only; every part takes one surface from a shared library of rule-based surfaces, written as ProcFunc functions [25] on Infinigen's shaders [9], coloured from the place's palette and worn only where wear has a cause (Appendix D). Gemini 3.1 Pro [26] lists the close-up's finishes and names a library surface for each. Nano Banana Pro draws the close-up again with one flat colour per finish, up to three times until the map fits the close-up's outline, then draws unlit renders of the raw model's sides and back against it. SegviGen [27] splits the mesh guided by the close-up's map, every map votes on the faces its view sees, and a part whose second finish holds a fifth of it fails. Each part is painted from its library surface and baked, never in the picture's colours, which gave every piece its own rust and stains (Appendix A).

The bake is checked against the close-up. A finish's surface must lie within ΔE 32 of its colour there, else a nearer surface of its family is taken or a new one is proposed to the creator, and the part fails by name. From the greenhouse's water tank, which the creator found baked as three near greys and tipped, came two checks: a bake fails when it darkens a finish the close-up shows lighter, or turns all surfaces into one grey; and short legs are stretched onto one level plane, a foot over 3 cm off the floor failing. Dirt follows the close-up.

3.2 Resting

Loose objects are dropped onto the made ground in a physics simulation; one that would turn more than 15 degrees or drift more than 30 cm is a resting fault of the layout and keeps its laid pose. The settle stage lays where each object comes to rest as a layer (held objects snapped onto their holders, a stack onto one support, an object that would rest worse kept as laid), gated on `rested` and `resting` (no fault that appears with the layer and not without it).

3.3 Light

Lamp strengths are measured, never copied from the source game. The light layer places live lamps from the scene record, with the sky, the sun and the Earth. One paid render per place gives every lamp its own Cycles light group on the place's views; because light adds up, the look at any strengths is mixed from these passes without rendering again, and a free sweep chooses the scale that brings the place's brightness within 1.5 times its concept's, halving any lamp that turns what it lights white. Outdoors the suns are swept after the lamps. A lamp whose fixture stops 90 percent of its rays moves out of it, and one behind a clear lens stands just outside, as Cycles passes no light through a glowing lens.

3.4 Motion, effects and sound

Motion. Each mover (a path, a hinge, a spin, a rail, a turn, a climb) has its kind's speed range cited from real machines, gated on `clear` (no mover within 3 cm of a still thing or another mover, read every quarter second), `speeds`, `loops`, `grounded` and `covered` (every still prim of a moving kind says why). **Effects** are emitters measured against real footage (photographs of lift-offs, rover films of Martian dust devils), gated on `attached`, `footage`, `light` (the flame lights the pad at least twice as brightly as the place's lamps) and `outside` (no open-sky particle inside a room). **Sound:** every source a place names (room tones, objects, motions, each surface's steps) gets the world's picked file

or the best of eight MOSS-SoundEffect [28] takes of its brief, gated on `sources`, `level`, `likeness` (CLAP [29] score at least 0.30, the tenth percentile of the world's picks) and `no freesound`.

3.5 People

People are made like props: a FLUX.2 klein [30] close-up in SOMA-X's bind pose, built by Pixal3D, painted by the surface line, over SOMA-X's body fitted just inside (Gauss-Newton, 24 shape numbers), whose skin weights each look point takes from its closest triangle; Kimodo's clips [31], from sentences, move it. The **shell gate** passes a look when 99 percent of its soft points lie within 5 cm of the body, its worst point recorded. Hard parts ride one joint, belts are cloth, each clip gets its own verdict (tucks, hard parts stretched, cloth or body through cloth) and casts use only passing clips; a far-only look is judged on its far body (Appendix B). **Casts:** SAM 3 [18] finds the concept's figures, a judge tells people from pictures, reflections and empty suits, and casts are gated on `counted`, `standing`, `clear`, `made` and `drapes`; no crowd copy fills a tenth of a view's height.

3.6 Completion

A place is complete when its gate passes on its final assembly: every row made or dropped with a reason, nothing unmade, no placeholder or resting fault, the creator's picks shown, every layer and shown place present, nothing cited from the source game. **Named gaps.** A fault the creator decides to ship with today is written down as a named gap: who decided, when, why, the objects and their measured faults. It moves only those objects' faults from the gate where they arise to the completion gate, which fails each gap by name with the numbers: a place whose named gaps are its only fail is complete apart from named gaps, every gap listed with who decided it; any other fault leaves it not complete, and the gate is never loosened. A cast's person with no accepted body is a named gap the same way. A hook stops an agent claiming a place is done while its gate fails: agents judging their own scenes agree with a measured geometry score only 45.8 percent of the time [15].

3.7 Budget and ledgers

Machines and Google calls write ledger lines with their node's ticket; the month's spend is the higher of ledgers and bill, which caught misses: EUR 1,554 billed against EUR 1,387 in the cloud ledger on 12 October 2026, and EUR 549 billed by Google for a day the earlier ledger put at EUR 30. Over 13.8 hours of world 1's run the ledgers hold EUR 117.45 of machines and EUR 34.19 of Google calls, most for painting (EUR 37.61), the finish maps (EUR 26.59), the people's bodies (EUR 16.09) and the light passes (EUR 10.17).

Forecast. Before a paid piece runs, its estimate is written down with the measured unit cost it rests on (the ledgers' medians per node, model or picture); after it, the ledger's actual. The projected total for the month is reported against a hard stop the creator sets. The record shows which estimates to trust: measured unit costs held (22 close-ups for the ship estimated at EUR 3.10 to 6.20, actual 3.50; seven box checks 1.75, actual 2.19; the ship's six rooms 27.40, actual about 16.70; its eight shell kits 1.20, actual 0.60), while the runner's flat table figure overstated work it could carry over (the prologue's surfaces and light, EUR 101.50 in the table, cost 0.86). The runner's forecast for a run is a ceiling, not an estimate: the four runs of 15 October were forecast at up to EUR 72.57, 33, 70.50 and 24 and cost 10.13, 10.14, 0.67 and 0.60, the last two stopped early because they ran older code. One ship run spent EUR 13.58 before it stopped on missing free inputs; the rule since then is a free pass over every input before a paid run. The bill runs about 21 percent over the ledgers, so the forecast multiplies every ledger-priced estimate by 1.21; on 15

October it put the month at EUR 495 to 507 against the creator's hard stop of EUR 510.

Held and run, 15 October. Before the night's runs the runner listed every stale node with its price. All people work was held for next month, EUR 89.50: 23 bodies (EUR 56.50, 6 to be made again after changes to the people maker, 17 not made yet), 12 takes (EUR 18), their maps, names, parts and paint (EUR 14) and one likeness calibration (EUR 1). Three review renders (EUR 14) were left out, as neither the completion gate nor the films read them. The place films, EUR 0.70 each, have not run.

4 Evidence

Four worlds are planned; world 1's 28 places are judged once their layers stand accepted.

Layouts from the source game. The framework's own layout, made from the plan and the concept, is proven only on the ship's 8 places and the 2 door kits. The other 18 places still take their layouts from the source game: 507 entries in their data cite the game, 338 of them positions, and 7 kits are laid in the game's frame, with 6,432 piece positions. The module rooms' shells, laid from the game's shell numbers, are not in these counts. The check sees cited numbers only, so an uncited copy passes: the launch's accepted scene record still lights the rocket at the game's position. Each of the 18 places ships this as a named gap of the completion gate's game check (Table 3); deriving their layouts again from plan and concept is a rebuild not yet priced.

People. Scored on what is drawn, no clip of any look is clean (Table 2). Takes in a wider A-pose crossed less at the bind and more in motion (10,610 visible crossings on the player's clips against 4,078), with no clip clean in either pose. People therefore ship as named gaps: the characters layer lays the people whose bodies stand accepted, and the completion gate fails every missing one by name.

Table 2. People, 13 October (experimental, Section 3.5): usable clips of 19 per look by the gate as built, on drawn surfaces only, and as one piece on what cameras see. Faults and fixes: Appendix B.

Look	Body gates	Gate	Visible	Cameras, one piece
Player (work)	pass	5	0	0
Bram (work)	pass	5	0	not run
Sefa (work)	pass	0	0	not run
Nev (work)	pass	0	0	not run
Kit, average man (olive)	fail: triangles	0	not run	not run
Player, wider pose (work)	fail: triangles	0	not run	0
Kit, average man, wider pose (olive)	fail: shell	0	not run	0

Table 3. Named gaps on 15 October: what world 1 ships without, failed by name in the completion gate or left out of this paper. The overseer, the coding-agent session that guards the run under the creator's rules, decided the gaps marked "to confirm" on the night of 15 October; they wait for the creator's confirmation.

Gap	Where	Why, and who decided	Next
Layouts from the source game	18 of 28 places: 507 citations, 338 positions; 7 kits in the game's frame, 6,432 piece positions	the framework's own layout is proven only on the ship's 8 places and the 2 door kits; overseer, to confirm	layouts from plan and concept, to be priced
People with no accepted body	every cast: the street's driver and guard, the square's leader, front row of 24 and crowd, two at the flat and two in the stairwell	no clip clean on what is drawn (Table 2); the creator's rule, 13 Oct; per place, overseer, to confirm	next month's people work
Likeness not checked	the prologue: street, launch, square, flat, stairwell	no props file, so generated models are not held to their close-ups; up to EUR 55 a place, outside world 1's budget; overseer, to confirm	props files
Cloth that stands still	the square's flags, banners and lanterns	cloth motion not built; overseer, to confirm	cloth motion
Two far towers	launch: the housing slab and the pencil tower	no stage builds the far city; overseer, to confirm	new code and the launch's prop data, about EUR 20
Finishes no library surface fits	cockpit: the throttle pedestal and the flight stick (5 finishes; four lie ΔE 25.6 to 34.7 from the nearest surface)	no red or orange plastic and no dark bare steel in the library; overseer, to confirm	new library surfaces, the creator's choice
A caster off the floor	lab: both desk chairs (4 cm)	the baked chair's caster sits above the others; overseer, to confirm	the model's feet
A foot off the floor	camp grounds' crane, greenhouse's seed tray	the level rule uses 1 percent, not 3 cm; overseer, 14 Oct	the same fix
The ship, code-built	hangar (12 pieces)	no made model yet; the creator, 14 Oct	the made ship, from the plan of Figure 1
Pieces that do not rest	wreck (5)	the settle drops models from before their bake; overseer, 14 Oct	the settle fix
Paint or bake off the close-up	garage (4), galley (1), navigation (1)	library surfaces held until next month; overseer, 14 Oct	new library surfaces
Results per place, place pictures	all	not made by this draft	after the place films
Pictures of the stages on the hub with its review page, of shape and surface apart, of light, and of things that move	hub, greenhouse, old station, launch, garage	not made by 07:30 on 15 Oct	after those places' completion gates
An edit that survives a regenerated base	all	no edit layer yet	an edit layer the assembly carries
Physics in an engine	one room	no engine adapter	an adapter
Worlds 2 to 4		not started	

The world 1 video. Each place is filmed once its completion gate passes or fails only on its named gaps: a 2 s title card, then one walk of 4 s (96 frames at 24 fps) on a path the planner keeps clear



Figure 3: The world 1 video (demo K9) on 18 October: the middle frame of each of its 27 films' walks, in story order. Every frame is the runner's accepted film of a place that is complete apart from named gaps; nothing was retouched or picked by hand.

of walls, objects and people, with the place's own sound. The film gate measures each film (its size, a sound track that is not silent, no black or frozen stretch after the title card, its length and the clearance) and the runner accepts it; no person reviewed the films before the cut, and the coding agent that made the cut checked only its order, frames and sound. One cut opens with the creator's concept for every place, then takes the places in story order, the prologue, the Moon base and the Moon, the ship, then Mars: each place's concept, plan and pieces where it has them, then 3.5 s of its accepted film's walk, with the ship's hull after the hangar; it closes on the layer stack of one place and what world 1 delivers for score_lib. On 18 October it held all 27 films, 26 places and the hull, 3 min 9 s (Figure 3). Four of them, the greenhouse, the garage, the hangar and the hull, were filmed again that day: the ground outside a joined tube stood up to 2 m inside those rooms, because the scene writer (tools/usd/scene.py, write_places) does not switch off a shown place's outside, and in the greenhouse it buried the crop bed; their film records now leave that ground out of the film's copy of the stage, and the fix in the scene writer is next month's. The airlock's film starts 6 s into its cycle, after its pump's fog: inside that fog a frame took from 47 s to 17 min on a card, and its first run, started in the fog, was stopped at the spend stop with its frames lost. Every filmed place was filmed after a completion gate that left it complete apart from named gaps. The final completion gate, run again on every place on 17 and 18 October, left all 26 places complete apart from named gaps (2 to 19 each) and none complete. The 2 door kits are not filmed; they are seen in the places they close.

Agent tools. With them an agent used about half the output and cost on two of three places; on the third it left more faults; one run each (Appendix E).

5 Limitations, future work and conclusion

One creator judges; the creator accepted the hub before the runner. Closed models (Nano Banana Pro, Gemini 3.1 Pro) may change or go. Hard belts failed deep bends; light is live, measured on earlier layouts; objects are not interactive; 18 of 28 places still take their layouts from the source game. Next: layouts from plan and concept in every place, three worlds, an engine adapter, interaction, baked lamps, open models, locked parts, LEGO-Bench [15]. In short, SCORE keeps, keys and judges every stage's output, so a fault is fixed where it happens; completed places will show if it holds.

References

[1] Y. Yang *et al.*, "Holodeck: Language guided generation of 3D embodied AI environments," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, 2024.

[2] D. Argüello, "36 percent of game workers use generative AI, but half think it's bad for the industry." Game Developer, Feb. 2026.

[3] J. Bruce, M. Dennis, A. Edwards, J. Parker-Holder, *et al.*, "Genie: Generative interactive environments," *arXiv preprint arXiv:2402.15391*, 2024.

[4] J. Parker-Holder and S. Fruchter, "Genie 3: A new frontier for world models." Google DeepMind blog, Aug. 2025. Available: https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/

[5] D. Wang, H. Jung, T. Monnier, K. Sohn, *et al.*, "WorldGen: From text to traversable and interactive 3D worlds," *arXiv preprint arXiv:2511.16825*, 2025.

[6] Tencent Hunyuan, "HY-World 2.0." GitHub repository, 2026. Available: https://github.com/Tencent-Hunyuan/HY-World-2.0

[7] J. Hu, J. Guo, J. Cen, C. Yang, S. Li, and W. Shen, "WorldAct: Activating monolithic 3D worlds into interactive-ready object-centric scenes," *arXiv preprint arXiv:2605.15843*, 2026.

[8] M. Niu, J. He, R. Yu, L. Fu, *et al.*, "WorldSculpt: Generating compositional worlds from grounded videos," *arXiv preprint arXiv:2609.05416*, 2026.

[9] A. Raistrick *et al.*, "Infinite photorealistic worlds using procedural generation," *arXiv preprint arXiv:2306.09310*, 2023.

[10] A. Raistrick *et al.*, "Infinigen Indoors: Photorealistic indoor scenes using procedural generation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, 2024.

[11] Z. Hu, A. Iscen, A. Jain, T. Kipf, *et al.*, "SceneCraft: An LLM agent for synthesizing 3D scene as Blender code," in *International conference on machine learning (ICML)*, 2024.

[12] Z. Li, Y. Liao, and B. Zhu, "Recursive code world models: Building complex worlds through recursive scene programs," *arXiv preprint arXiv:2609.11499*, 2026.

[13] C. Guo, J. Li, Y. Li, and Z. Huang, "WorldClaw: Agentic 3D open-world generation at scale," *arXiv preprint arXiv:2608.05248*, 2026.

[14] L. Yin, W. Cheng, Z. Qin, T. Huang, *et al.*, "AutoUE: Automated generation of 3D games in Unreal Engine via multi-agent systems," *arXiv preprint arXiv:2603.07106*, 2026.

[15] X. Li *et al.*, "LEGO-Anything: Coding agents for 3D scene reconstruction," *arXiv preprint arXiv:2609.36380*, 2026.

[16] M. Oquab, T. Darcet, T. Moutakanni, *et al.*, "DINOv2: Learning robust visual features without supervision," *arXiv preprint arXiv:2304.07193*, 2023.

[17] D.-Y. Li *et al.*, "Pixal3D: Pixel-aligned 3D generation from images," *arXiv preprint arXiv:2605.10922*, 2026.

[18] N. Carion *et al.*, "SAM 3: Segment Anything with concepts," *arXiv preprint arXiv:2511.16719*, 2025.

[19] R. Wang *et al.*, "MoGe-2: Accurate monocular geometry with metric scale and sharp details," *arXiv preprint arXiv:2507.02546*, 2025.

[20] Y. Lin *et al.*, "PartCrafter: Structured 3D mesh generation via compositional latent diffusion transformers," in *Advances in neural information processing systems (NeurIPS)*, 2025.

[21] J. Wu, Z. Lou, J. Liu, D. Du, C. Guo, and S. Guo, "From visual synthesis to interactive worlds: Toward production-ready 3D asset generation," *arXiv preprint arXiv:2604.23629*, 2026.

[22] Z. Yang *et al.*, "FlowScene: Style-consistent indoor scene generation with multimodal graph rectified flow," *arXiv preprint arXiv:2603.19598*, 2026.

[23] Pixar Animation Studios and Alliance for OpenUSD, "OpenUSD: Universal scene description." 2026. Available: https://openusd.org/

[24] Google, "Gemini 3 Pro Image ('Nano Banana Pro') through the Gemini API." 2026. Available: https://ai.google.dev/

[25] A. Raistrick *et al.*, "ProcFunc: Function-oriented abstractions for procedural 3D generation in Python," *arXiv preprint arXiv:2604.26943*, 2026.

[26] Google, "Gemini 3.1 Pro (gemini-3.1-pro-preview) through the gemini API." 2026. Available: https://ai.google.dev/

[27] L. Li *et al.*, "SegviGen: Repurposing 3D generative model for part segmentation," *arXiv preprint arXiv:2603.16869*, 2026.

[28] OpenMOSS-Team, "MOSS-SoundEffect v2.0." Hugging Face, 2026. Available: https://huggingface.co/OpenMOSS-Team/MOSS-SoundEffect-v2.0

[29] Y. Wu, K. Chen, T. Zhang, *et al.*, "Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation," *arXiv preprint arXiv:2211.06687*, 2022.

[30] Black Forest Labs, "FLUX.2 [klein] 4B model card." Hugging Face, 2026. Available: https://huggingface.co/black-forest-labs/FLUX.2-klein-4B

[31] D. Rempe, M. Petrovich, Y. Yuan, H. Zhang, X. B. Peng, *et al.*, "Kimodo: Scaling controllable human motion generation," *arXiv preprint arXiv:2603.15546*, 2026.

[32] M. Stone, D. A. Szafir, and V. Setlur, "An Engineering Model for Color Difference as a Function of Size," in *Color and imaging conference (CIC 22)*, Society for Imaging Science; Technology, 2014. Available: https://library.imaging.org/admin/apis/public/api/ist/website/downloadArticle/cic/22/1/art00045

[33] Blender Foundation, "Blender." 2026. Available: https://www.blender.org/

[34] Qwen Team, "Qwen-Image-Edit-2511 model card." Hugging Face, 2025. Available: https://huggingface.co/Qwen/Qwen-Image-Edit-2511

[35] Qwen Team, "Qwen3.8-27B model card." Hugging Face, 2026. Available: https://huggingface.co/Qwen/Qwen3.8-27B

[36] O. Siméoni, H. V. Vo, M. Seitzer, *et al.*, "DINOv3," *arXiv preprint arXiv:2508.10104*, 2025.

[37] K. Deng *et al.*, "GeoSAM2: Unleashing the power of SAM2 for 3D part segmentation," *arXiv preprint arXiv:2508.14036*, 2025.

[38] L. Yang, B. Kang, Z. Huang, *et al.*, "Depth Anything V2," *arXiv preprint arXiv:2406.09414*, 2024.

[39] World Labs, "Terms of Service." 2026. Available: https://docs.worldlabs.ai/terms-of-service

[40] P. BehnamGhader, V. Adlakha, M. Mosbach, D. Bahdanau, N. Chapados, and S. Reddy, "LLM2Vec: Large language models are secretly powerful text encoders," in *Conference on language modeling (COLM)*, 2024.

[41] A. Grattafiori, A. Dubey, A. Jauhri, *et al.*, "The Llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.

[42] X. Yang, D. Kukreja, D. Pinkus, others, and K. Kitani, "SAM 3D Body: Robust full-body human mesh recovery," *arXiv preprint arXiv:2602.15989*, 2026.

[43] A. Qammaz, "SAM3DBody-cpp: A standalone C++ inference engine for SAM 3D Body." GitHub, 2026. Available: https://github.com/AmmarkoV/SAM3DBody-cpp

[44] M. Korosteleva and O. Sorkine-Hornung, "GarmentCode: Programming parametric sewing patterns," *ACM Transactions on Graphics*, vol. 42, no. 6, p. 197, 2023.

[45] The Newton Developers, "Newton: A GPU-accelerated physics simulation engine." 2026. Available: https://github.com/newton-physics/newton

[46] M. Macklin, "Warp: A high-performance Python framework for GPU simulation and graphics." NVIDIA, 2022. Available: https://github.com/NVIDIA/warp

[47] C. Ye *et al.*, "Hi3DGen: High-fidelity 3D geometry generation from images via normal bridging," in *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)*, 2025.

[48] Black Forest Labs, "FLUX.2 [klein] Base 4B model card." Hugging Face, 2026. Available: https://huggingface.co/black-forest-labs/FLUX.2-klein-base-4B

[49] thedeoxen, "refcontrol: Reference-depth LoRA for FLUX.2 [klein] 4B." Hugging Face, 2026. Available: https://huggingface.co/thedeoxen/refcontrol-FLUX.2-klein-4B-reference-depth-lora

[50] J.-P. Zhang, C.-F. Pu, M.-H. Guo, Y.-P. Cao, and S.-M. Hu, "One model to rig them all: Diverse skeleton rigging with UniRig," *ACM Transactions on Graphics (SIGGRAPH)*, 2025.

[51] R. Abdrashitov, K. Raichstat, J. Monsen, and D. Hill, "Robust skin weights transfer via weight inpainting," in *SIGGRAPH asia 2023 technical communications*, 2023.

[52] Anthropic, "Claude Code." 2026. Available: https://www.anthropic.com/claude-code

[53] H. I. I. Tam, H. I. D. Pun, A. T. Wang, A. X. Chang, and M. Savva, "SceneEval: Evaluating semantic coherence in text-conditioned 3D indoor scene synthesis," *arXiv preprint arXiv:2503.14756*, 2025.

[54] H. Xia *et al.*, "SAGE: Scalable agentic 3D scene generation for embodied AI," *arXiv preprint arXiv:2602.10116*, 2026.

Appendix

A The hub in six rounds

The surface library, one model per real-world object and the straightness check came out of six rounds of work on the hub on 6 and 7 October 2026, each answering the creator's verdict on the round before (Table A1). In the first two rounds, the creator chose between the old and the new method side by side from the same cameras.

Table A1. The hub's six rounds. Cost is given as rented GPU time in euros and picture-model calls in dollars. Frame time is the mean frame time on an RTX 5080, in the night scene, at 1600×900.

Round	Change	Creator's verdict	Cost	Frame time
1, two walls, A/B	surface library, routing by shape, room checks	chose the new method ("Y looks much cleaner")	€0.88	7.81 ms vs 7.53
2, two walls, A/B	71-variant library, shared texture sets, roof check	chose the new method again	€1.70	8.05 ms vs 8.24
3, whole room	all stages end to end, 74 models	old models were still loaded; labels and floor fittings were wrong	€1.67	6.78 ms (old 9.22)
4, whole room	detail through the pipeline, picture colours kept	"pretty much all of this became messy"	€8.50, $5.09	9.6 ms
5, eight pieces, A/B	shape only from the pipeline, library surfaces, parts check	new method preferred for the door, notice board and porthole; three pieces still wrong	€0.21	not run
6, whole room	round 5 everywhere, one model per object, straightness check	"this looks amazing, I have no negative feedback"	€3.48, $1.07	5.6 ms

Three findings shaped the method. First, deterministic checks catch what agents miss: in round one, the share of rays leaking out through walls and roof fell from 1.53 percent of 5,817 to zero, and the colour noise within surfaces from a median of 31.1 to 3.2 percent, but only walking through the loaded room showed 26 old models still in it, which became the made-only check. Second, generated pieces that kept their pictures' colours made the room inconsistent, so surfaces come only from the library. Third, generated composite objects melted their parts together, and generated boxes leaned (one locker by 13 degrees), which led to one model per real-world object and to the straightness check. By round six the room used 0.45 million triangles, against 1.96 million in round four, and 187 MB on disk, against 487 MB.

B Check details

- **Room checks** cast rays to find light leaking out through walls and roof, find pieces in front of a real opening and roof and wall fittings more than 3 cm off their surface or turned more than 5 degrees from it, and count pieces, triangles and lamps against the room's budget.
- **Door check:** every door must stand in a wall or partition that fully separates its two sides, tested by walking a grid from one side to the other.
- **Model check:** every model must have walls at least 3 mm thick. A generated box fails if one of its sides tilts more than 2 degrees or is warped. Making a generated model solid stops if its bounding box grows more than 3 percent.
- **Likeness gate:** a made model's outline from eight sides must overlap its close-up's by at least 0.50, a least set on the hub's twelve accepted generated models with a margin of 0.05, and its DINOv2 [16] similarity must reach a least set the same way; without that calibration the gate answers unknown.
- **Finish maps:** a close-up's map passes when its outline overlaps the close-up's by at least 0.85 and it uses every listed finish; a part whose second finish holds 20 percent of its voted area mixes finishes and fails.
- **Bake check:** each baked part within ΔE 20 (CIE76) of its surface's colour, worn and dirtied as far as its place's wear and dirt, and within ΔE 32 of its close-up's; lightness order fails when the close-up shows one surface more than 10 L* lighter and the bake makes it more than 3 L* darker; one look fails when no two near-neutral surfaces bake far enough apart for a viewer to tell at the place's view distance, while the close-up shows them more than ΔE 24 apart. What a viewer tells is measured, not set: Stone, Szafir and Setlur measured the CIELAB difference half of their viewers see between two colours s degrees wide, per axis ND(50, s) = C + K/s, with (C, K) = (5.08, 0.75) for L*, (5.34, 1.54) for a* and (5.35, 2.87) for b* on sizes 1/3 to 6 degrees [32]; two surfaces are told apart when their difference, each axis over its ND, is 1 or more. A surface's size is its model's middle size times the root of its area share, seen from the place's nearest view. This replaced a fixed ΔE 12, which failed the wreck's access cover on a 9.3 L* step between its steel and its scorched white that its owner sees (11.3 degrees from the nearest view; half of viewers see 5.2 L* at that size); a quadrant of a standing model's base more than 3 cm off the floor fails. Glass, screens, lamps and print are judged by their own colour and left out of the order.
- **Settling and resting check:** an object that would turn more than 15 degrees or drift more than 30 cm when dropped keeps its laid pose and is marked as not resting; small debris, with no side over 1.4 m, may turn freely and drift up to 1 m. Hung objects are exempt by their inventory anchor, and fixed objects are not judged for tipping. The triage fails a foot off its support and an up axis leaning more than 3 degrees.
- **Light:** the place's median brightness within 1.5 times its concept's and every view within 2.5 times; a lamp is blown when over 20 percent of the pixels where it gives at least half the light are white, and unlit when it gives less than 0.001 percent of every view's light.
- **Motion:** no mover within 3 cm of a still thing or another mover, read every 0.25 s; rounds join within 1 mm; speeds within each kind's cited range.
- **Sound:** CLAP score against the brief at least 0.30, the tenth percentile of 55 of the world's picks; a take no more than 0.5 s longer than its brief.
- **People:** the shell gate passes a look when 99 percent of its soft points lie within 5 cm of the fitted body; the worst point and its joint are recorded. A clip is usable when no frame tucks more than 5 mm, no hard part stretches past 1.2 times its bind length, cloth crosses cloth no more than in the earlier body's same clip, and no part's edges pass through cloth deeper than

3 mm more often than at the bind. A far-only look is judged on its joins, likeness and the far body's triangles (at most 1,220); the near outfit's triangles and shell offset are recorded, not gating. The A-pose outline is judged against the finished Pixal3D take the build starts from; the take's own loss against the raw model is recorded apart. A crowd copy stands only where it fills under a tenth of every view's height.

- **Worse than accepted:** a measure counts as worse only when it moves the wrong way by at least its tolerance, the smallest change a viewer could see: one more fault for a count; 0.02 of full scale for lightness on 0 to 1 (a halo, fill, beam or haze wash), and 5 of 255 levels for a colour gap, the step a viewer notices between two pictures; 0.05 for a brightness ratio; one percent of the picture for a share of it; a twentieth of the Earth's lit disc; 5 mm for a length. A score no viewer reads directly, such as a likeness score, is worse by any move. Tolerances decide only whether an output waits for the creator; a check that fails still fails.
- **The creator's accept carried forward:** when the creator accepts a worse output, that accept records what was allowed, each measure's value before and after. A later output that passes every gate and is no worse than the output taken, within the tolerances, is accepted by the runner with that accept named; a move beyond it still waits for the creator.
- **What the creator acts on:** the creator rejects, accepts or waives only an output already seen on its review page. This is the creator's rule; the commands do not yet check that the page was opened.
- **People seen by the cameras (experimental):** a clip's crossings of body through cloth are counted only where a camera that will show the person can see them: the place's views at the person's cast spot, and a close front and three-quarter view. Crossings no camera sees are recorded, not counted against the clip. So far this is a measure on the stored bodies (Section 3.5); the per-clip gate does not use it yet.
- **Completion gate:** the requirements given in Section 3.6, each with its result tied to a fingerprint of the stage's files and inputs; a result made on another stage counts as unknown. The placeholder check judges every mesh a code builder draws, and only the shell builders pass unnamed.

People results in detail (experimental, Section 3.5). The body gates are joins, outline, likeness, triangles and shell. Nev's and Sefa's boot tuck bands lie 9 to 11 mm outside the trouser leg at the bind pose, in every clip before any motion. Scored on drawn surfaces only, no clip of any look is usable: skin seams open and settled cloth crosses itself beyond the bind. Bound as one closed mesh, the player's limbs sink into chest and hips, 87 to 563 crossing points visible from the places' cameras and close views per clip, the armpit surface stretching between arm and chest. Where hair or cloth meets the head or hand skin the two sides are weighted differently and open as the joints move (cloth seams never open); openings over 10 cm on three looks are not yet confirmed by a render. Binding the take as one closed mesh removes the openings. Pushing points out of the look's own volume, limiting the offset from SOMA-X at the armpits and inner thighs, fold-centred weights and dual-quaternion skinning each left the visible crossings the same or higher. A close-up drawn in a wider A-pose (arms 45 degrees below level, feet 35 cm apart) gives a take with fewer crossings at the bind (the player's 678 against 772) but more in motion, as the clips bring the arms further down: 10,610 visible crossings against 4,078 over the player's 19 clips, and no clip clean in either pose. The clips' joint angles stay within the normal human ranges of the American Academy of Orthopaedic Surgeons, except hip extension of 24 and 27 degrees (limit 20) in two clips.

Decisions recorded in these checks. Three people's gates are decisions of the captain, the agent session that runs world 1, taken on 12 and 13 October 2026 under the creator's rules and open to the creator's reversal. (1) The shell gate counts 99 percent of the soft points rather than

every point, with the worst point recorded: the points beyond it were loose cloth edges (cuffs, hems, caps), which no part label names, so they could not be given a joint of their own by label. (2) A look seen only far away is judged on what is drawn far away, not on the near outfit it was made from. (3) The A-pose outline is judged against the finished take, not the raw model, so the take's own loss is not charged to the build. Earlier, the full-body cut (which looks get bodies in full and which only far ones) was set by the people's share of the budget.

C Models and licences

Every model in the framework must allow commercial use of its output (Table C1); we read each licence for its clauses on commercial use. Candidates were dropped under this rule, among them NVIDIA's Lyra 2.0 (weights for internal research only), Hunyuan3D 2.1 (licence does not apply in the European Union) and GarmentCode's fork of NVIDIA Warp (non-commercial research only), which Newton's cloth solver on upstream Warp replaced; Blender's cloth simulation [33] is kept as a second route. The one open doubt is the face LoRA, whose training data is not disclosed.

Table C1. Models in the framework.

Model	Role	Licence
Nano Banana Pro [24]	concepts, close-ups on the prop route and those the open model fails, ground skins	paid API
Nano Banana Pro [24]	finish maps: the close-up drawn again with one flat colour per finish	paid API
Gemini 3.1 Pro [26]	each model's finishes listed on its close-up, and their library surfaces named	paid API
Qwen-Image-Edit-2511 [34]	close-ups, first (before the prop route)	Apache-2.0
Qwen3.8-27B [35]	the close-ups' shape check; each part's material	Apache-2.0
Pixal3D [17]	picture to 3D	MIT; DINOv3 licence [36] for its encoder
PartCrafter [20]	parts	MIT; its non-commercial background remover not used
SegviGen [27]	lays the close-up's finish map onto the 3D shape (surface stage)	MIT, on TRELLIS.2 (MIT); DINOv3 licence for its encoder; its non-commercial background remover and nvdiffrast not used
GeoSAM2 [37]	parts of our own model, seeded by the close-up's regions (trial)	Apache-2.0
Depth Anything V2 Small [38]	ground relief	Apache-2.0
ProcFunc, Infinigen shaders	surfaces	BSD-3-Clause
MOSS-SoundEffect v2.0, CLAP	sound	Apache-2.0
World Labs [39]	optional world step	paid; outputs owned by paid users
FLUX.2 klein 4B [30]	earlier prop pictures; the characters' A-pose pictures from words and their hair's clay pictures	Apache-2.0
MoGe-2 [19], SAM 3 [18]	measuring and finding objects in pictures; counting people and crops on the concepts	MIT; SAM Licence
Kimodo [31]	the characters' motion from sentences	Apache-2.0 code; NVIDIA Open Model License for the Kimodo-SOMA-RP-v1.1 weights, commercial use allowed

Model	Role	Licence
LLM2Vec [40] on Meta-Llama-3-8B-Instruct [41]	Kimodo's text encoder, used only while clips are made; nothing from it ships	Llama 3 Community License: commercial use allowed, the 700 million monthly users clause does not apply, "Built with Meta Llama 3" shown, no exclusion in the European Union for the text-only 8B model
SAM 3D Body [42], SAM3DBody-cpp [43]	body shape from a picture	SAM Licence for the weights, SAM3DBody-cpp's converted ONNX and GGUF weights included whatever their tag says; MIT for SAM3DBody-cpp's code
SOMA-X and MHR	the characters' bodies	Apache-2.0
GarmentCode [44], Newton [45] on Warp [46], Blender's cloth [33]	the clothes' sewing patterns and their drape	MIT; Newton and Warp Apache-2.0; Blender is GPL and its output is ours; GarmentCode's Warp fork (non-commercial research only) not used
Hi3DGen [47]	the characters' hair	MIT code; MIT and Apache-2.0 weights
FLUX.2 klein base 4B [48] with the refcontrol reference-depth LoRA [49]	the characters' faces	Apache-2.0, both; the LoRA's training data is not disclosed
UniRig [50]	the skeleton and skin of a quadruped	MIT code and weights
Robust Skin Weights Transfer [51]	the SOMA-X skeleton's skin weights moved onto a Pixal3D person (a trial)	MIT reference code, ported with trimesh and SciPy; libigl, whose bindings carry GPL-3.0, not used
MakeHuman eyes	the characters' eyes	CC0

The close-ups' shape check was tuned on 40 close-ups that four open picture models drew of the lab's ten objects, each scored by hand, and on Nano Banana Pro's ten pictures of the same objects. With the majority of three sampled answers, the check agreed with the hand score on 29 of the 40, passed all ten of the closed model's pictures and two marginal ones, passed none scored wrong or failed, and failed nine that were good or usable. A single answer at temperature zero was rejected: asked 51 questions twice in one batch, it changed 7 verdicts. On the lab, 7 of the closed model's first pictures drew the room or dimension lines; all 7 passed when redrawn from the crop alone.

Table C2. The close-up stage on world 1. Cloud is rented card time in euros; the closed model's pictures are in dollars.

Run	Close-ups	Open model accepted	Closed model accepted	Left for the creator	Cloud	Closed model
Lab, 22 rows (earlier check: one answer)	22	8	14	0	€4.12	$2.81
World 1's remaining rows (three places)	17	0	10	7	€3.06	$4.02

The picture-to-3D step, the part splitting, the surface bakes, the sound generation, all checks and the assembly run automatically. Coding agents (Claude Code [52]) write the plans, the inventories and every code builder; in the hub's final round, 86 of its 98 models were built in code. Some steps were still done by hand: some generated objects were turned to face the right way by eye, and a

few labels were placed by hand before a rule took over that job. The concepts and the verdicts are the creator's own choices.

D The surface library

The surface library holds families of variants, among them painted, steel, aluminium, rubber, plastic, cable, fabric, composite, glass, screen, light, print, wood, ground, foil, water, skin and hair. After the hub's second round it held 71 variants made from 14 ProcFunc recipes; it now holds 369 variants in 26 families made from 22 recipes. Each variant is a recipe with settings, takes its colours only from the palette, and is baked into base-colour, metal-roughness and normal maps at a texture density that depends on how close the player can get. Wear has three levels and comes from causes: edges, and kicks within 32 cm of the floor that a piece stands on. A surface the finish-map route proposes enters an append-only registry and joins the library only when the creator accepts it.

E With and without the agent tools

The tools that check an agent's work were tested on three places of world 1: a room (the workshop, 814 objects), an outdoor place (the street, 1,198 objects) and a dense room (the hangar, 1,657 objects). They are the completion gate (Section 3.6); a triage of the resting check, which judges each object by what holds it up, read from its inventory row, and keeps only real faults [53], [54]; a pick lock, which refuses an export that would show anything but the model the creator picked for a row; and a render-and-compare, which lists the objects least like their close-ups by DINOv2 similarity [16]. In both arms, the same agent (Claude Opus 5.5 in Claude Code, headless) started from the same exported stage and inventory and was asked to make the place as complete as it could in about 50 minutes, through its layout data only; one arm had the framework from before these tools. Each arm ran once, so Table E1 shows what happened, not an average.

Table E1. One session per arm and place. Faults are the resting triage's real faults left on the final stage, measured against the arm's own data; placeholders by the first placeholder check; rows are inventory rows with no object and no written reason. Agent cost is the session's model cost; cloud is its rented machines' cost.

Place, arm	Faults, start → end	Placeholders	Rows missing	Min.	Agent cost	Output tokens	Cloud	Exports	Unknown gate calls
Workshop, without	18 → 13	0 → 0	1 → 0	36	$2.26	30.4k	€0.85	7	not used
Workshop, with	18 → 0	0 → 0	1 → 0	26	$1.42	14.3k	€0.85	2	2 of 4
Street, without	11 → 0	0 → 0	0 → 0	38	$2.40	30.0k	€0.32	9	not used
Street, with	12 → 0	0 → 0	0 → 0	39	$1.21	10.4k	€0.76	1	0 of 5
Hangar, without	14 → 4	5 → 0	0 → 0	47	$2.01	26.6k	€0.00	7	not used
Hangar, with	14 → 9	5 → 0	0 → 0	32	$2.34	29.1k	€0.64	5	3 of 6

The tools made building cheaper more often than they made the scene better. On the workshop and the street, the arm with the tools used about half the agent's output and cost, and exported the stage once or twice instead of seven to nine times, because the triage named the few real faults among hundreds of alarms. On the workshop, 13 of its 18 faults were cleared by writing into the inventory what holds a piece up, which the creator still has to confirm. On the hangar the arm without the tools left fewer faults: the arm with them spent its time replacing five placeholder slabs

and trying the drop test, which failed on a room with no ground mesh (fixed after the run), and the nine faults it left are models with stray parts below their feet. No agent claimed a place was complete, so the gate's hook never had to refuse a claim. No row carries an owner's pick, so the pick lock was not tested.